

```
from google.colab import files
```

```
uploaded = files.upload()
```

[Choose Files](#) placement_...Dataset.csv

placement_predict_50k Dataset.csv(text/csv) - 6811359 bytes, last modified: 8/4/2026 - 100% done
Saving placement_predict_50k Dataset.csv to placement_predict_50k Dataset.csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.backends.backend_pdf import PdfPages

sns.set_style("whitegrid") # optional: gives cleaner-looking charts
```

```
df = pd.read_csv("placement_predict_50k Dataset.csv")
```

```
print("First 5 rows of the dataset:")
display(df.head()) # use display() in Colab; use print(df.head()) elsewhere
```

First 5 rows of the dataset:

	StudentID	Gender	City	CollegeTier	Stream	Specialisation	Hostel	HistoryOfBacklogs	SGPA_Sem1	SGPA_Sem2	...
0	1	Male	Ahmedabad	Tier2	ECE	Networking	No	No	6.02	6.54	...
1	2	Female	Mumbai	Tier2	ECE	DataScience	Yes	Yes	5.84	5.12	...
2	3	Male	Kolkata	Tier2	IT	DataScience	Yes	No	4.91	5.29	...
3	4	Male	Jaipur	Tier1	CS	AI	No	No	7.67	8.03	...
4	5	Male	Pune	Tier2	IT	DataScience	Yes	No	8.14	8.97	...

5 rows × 32 columns

```
print("\nShape of dataset (rows, columns):", df.shape)
```

```
print("\nColumn names:")
print(df.columns.tolist())
```

```
print("\nData types and non-null counts:")
df.info()
```

```
print("\nMissing values per column:")
print(df.isnull().sum())
```

```
11 SGPA_Sem4      50000 non-null float64
12 SGPA_Sem5      50000 non-null float64
13 SGPA_Sem6      50000 non-null float64
14 SGPA_Sem7      50000 non-null float64
```

```

CollegeTier      0
Stream           0
Specialisation   0
Hostel           0
HistoryOfBacklogs 0
SGPA_Sem1        0
SGPA_Sem2        0
SGPA_Sem3        0
SGPA_Sem4        0
SGPA_Sem5        0
SGPA_Sem6        0
SGPA_Sem7        0
SGPA_Sem8        0
CGPA             0
AttendancePercent 0
Internships      0
Projects         0
Workshops        4488
Certifications   0
Publications     0
AptitudeTestScore 4029
SoftSkillsRating 3526
CodingTestScore  2976
MockInterviewScore 4957
ExtraCurricular 0
CGPA_Tier        0
PlacementStatus  0
IsAnomaly        0
Salary Package   0
dtype: int64

```

```
df['Placed_Flag'] = df['PlacementStatus'].apply(lambda x: 1 if x == 'Placed' else 0)
```

```

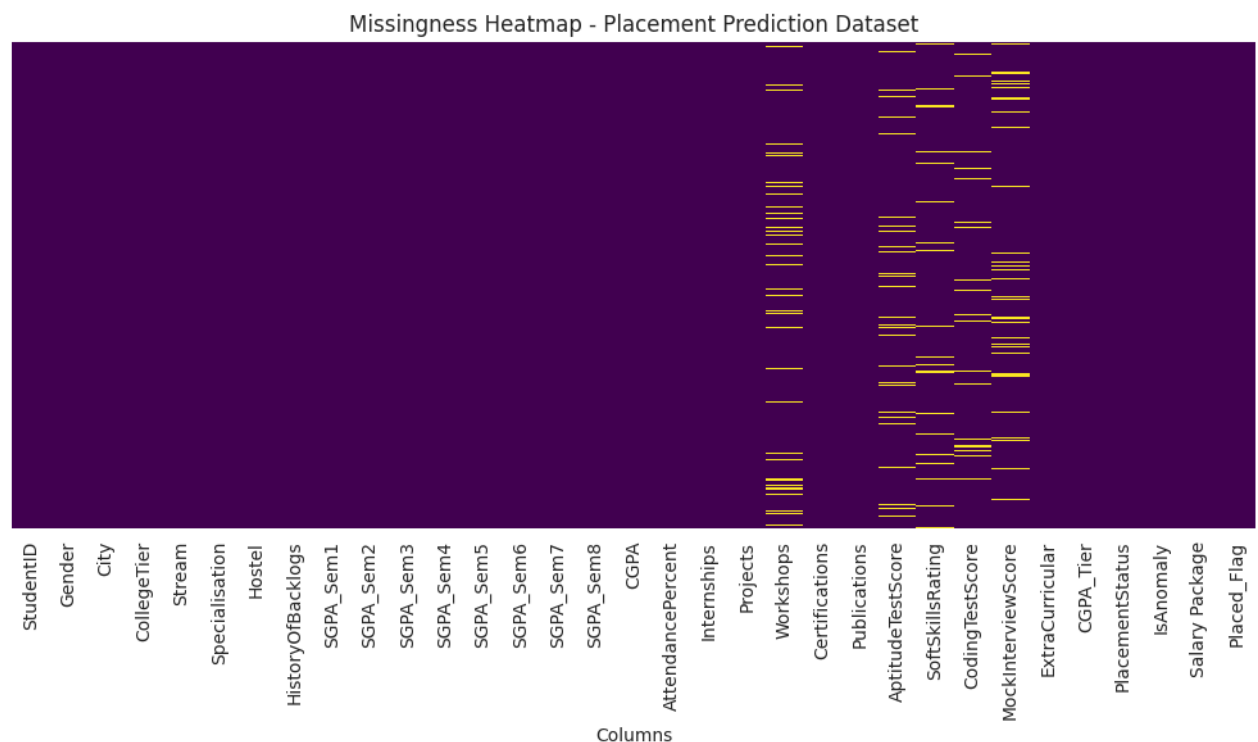
num_cols = ['CGPA', 'AttendancePercent', 'AptitudeTestScore',
            'CodingTestScore', 'SoftSkillsRating', 'Salary Package']

```

```

fig1 = plt.figure(figsize=(10, 6))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis", yticklabels=False)
plt.title("Missingness Heatmap - Placement Prediction Dataset")
plt.xlabel("Columns")
plt.tight_layout()
plt.show()

```



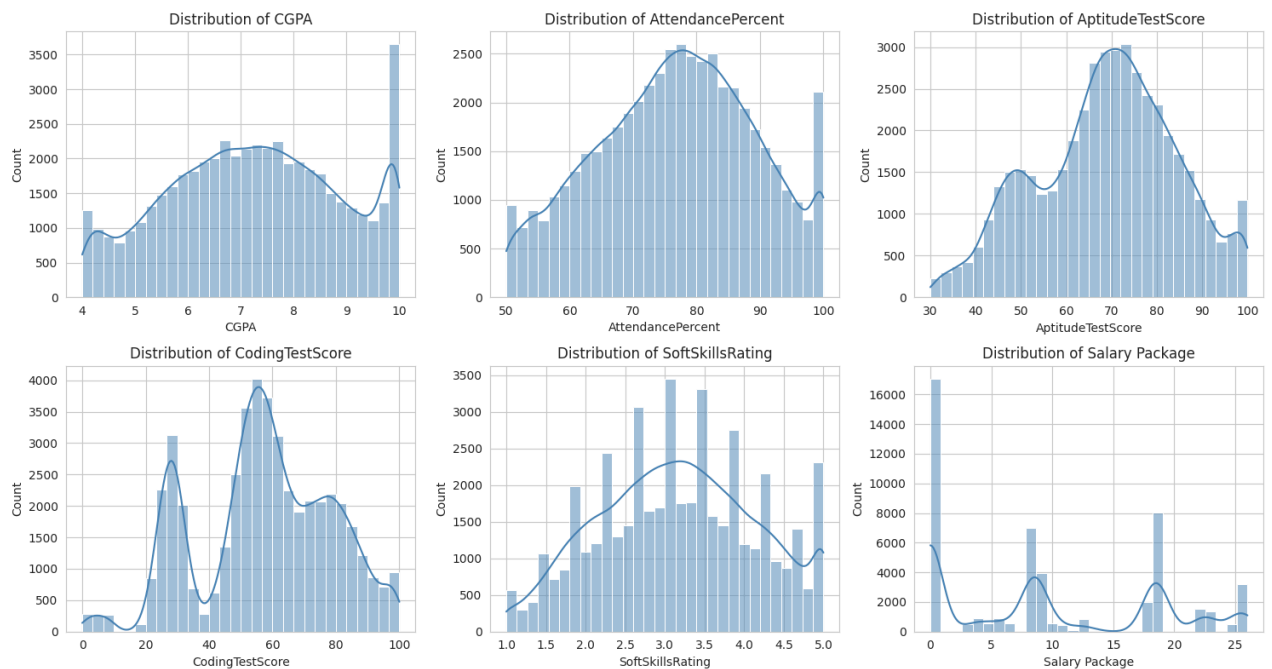
```
fig2, axes = plt.subplots(2, 3, figsize=(15, 8))
```

```

for i, col in enumerate(num_cols):
    row, colpos = divmod(i, 3)
    sns.histplot(df[col].dropna(), bins=30, kde=True,
                 ax=axes[row, colpos], color='steelblue')
    axes[row, colpos].set_title(f'Distribution of {col}')

plt.tight_layout()
plt.show()

```



```

fig3 = plt.figure(figsize=(10, 6))

placement_rate = (
    df.groupby('Stream')['Placed_Flag'].mean().sort_values(ascending=False) * 100
)

sns.barplot(x=placement_rate.index, y=placement_rate.values, palette='crest')
plt.title('Placement Rate (%) by Stream')
plt.xlabel('Stream')
plt.ylabel('Placement Rate (%)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

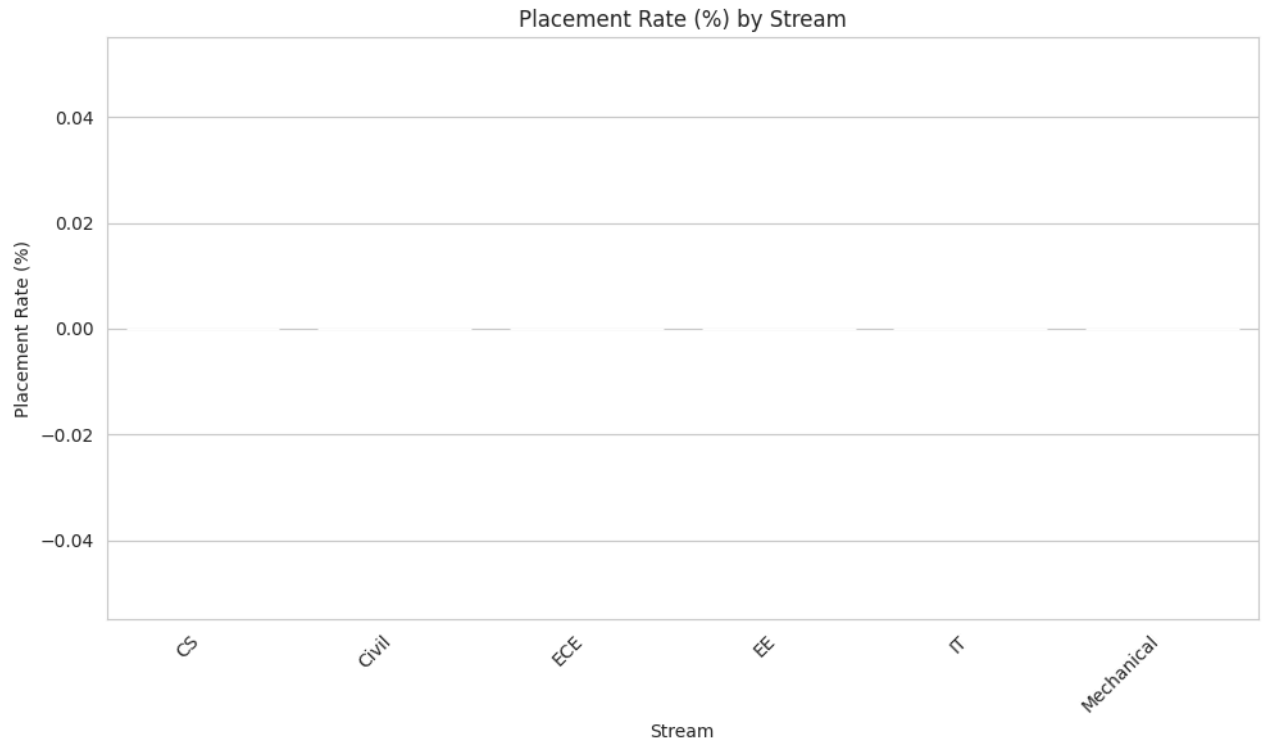
print("\nPlacement rate (%) by stream:")
print(placement_rate.round(2))

```

/tmp/ipykernel_485/2158546150.py:7: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.barplot(x=placement_rate.index, y=placement_rate.values, palette='crest')
```



Placement rate (%) by stream:

Stream

CS 0.0

Civil 0.0

ECE 0.0

EE 0.0

IT 0.0

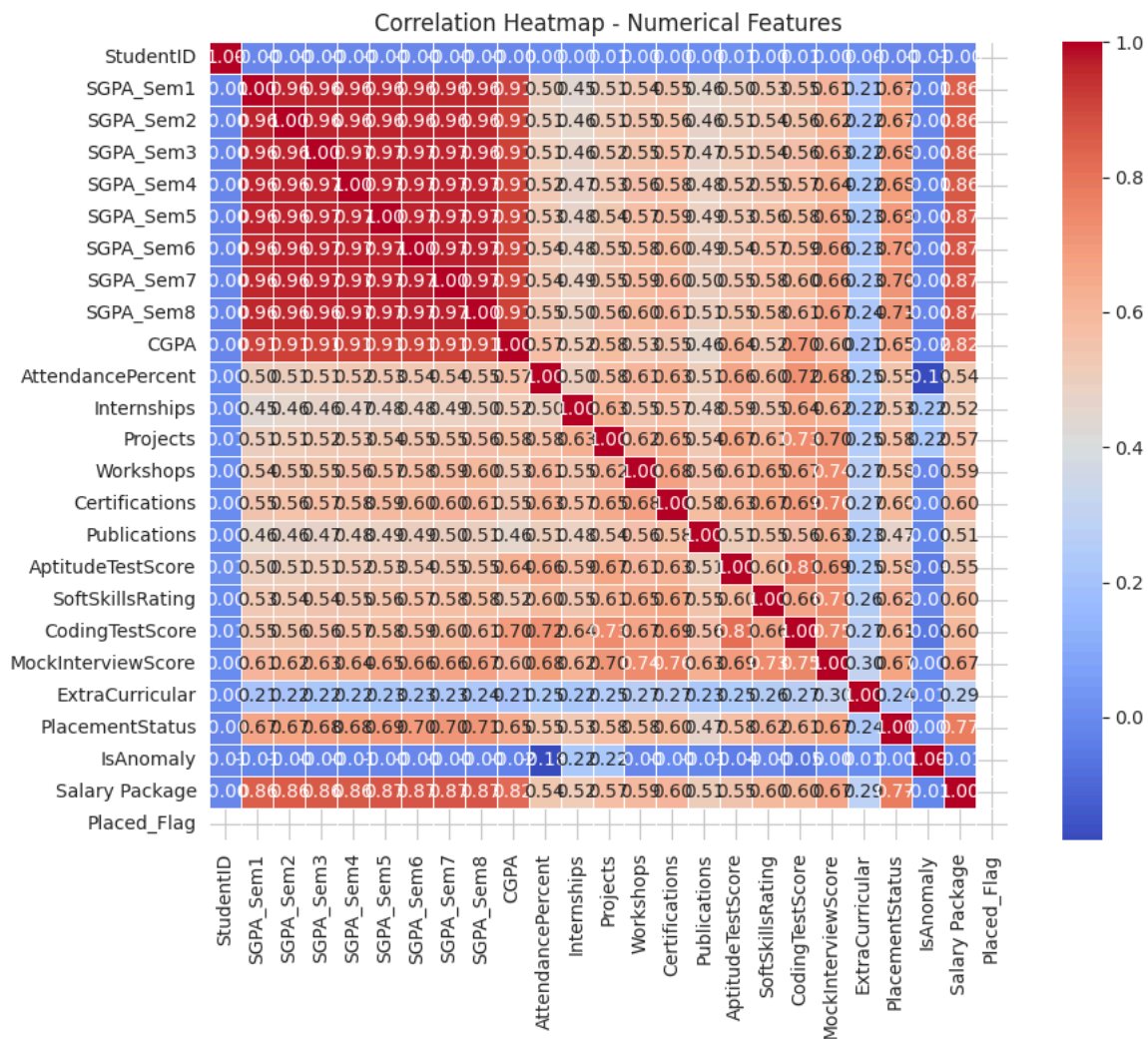
Mechanical 0.0

Name: Placed_Flag, dtype: float64

```
fig4 = plt.figure(figsize=(10, 8))

numeric_df = df.select_dtypes(include=['int64', 'float64'])
corr_matrix = numeric_df.corr()

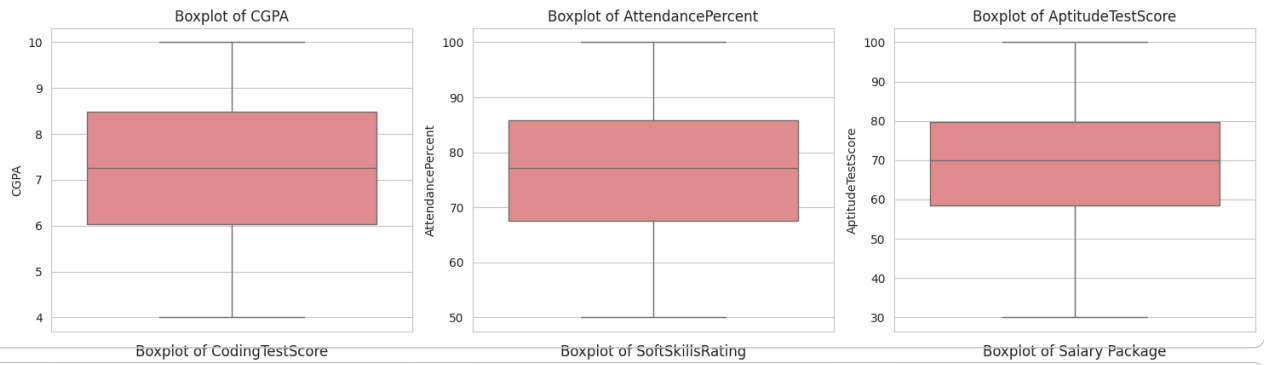
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            square=True, linewidths=0.5)
plt.title('Correlation Heatmap - Numerical Features')
plt.tight_layout()
plt.show()
```



```
fig5, axes5 = plt.subplots(2, 3, figsize=(15, 8))

for i, col in enumerate(num_cols):
    row, colpos = divmod(i, 3)
    sns.boxplot(y=df[col], ax=axes5[row, colpos], color='lightcoral')
    axes5[row, colpos].set_title(f'Boxplot of {col}')

plt.tight_layout()
plt.show()
```



```
pdf = PdfPages('EDA_Report.pdf')
```

```
pdf.savefig(fig1) # Page 1: Missingness Heatmap  
pdf.savefig(fig2) # Page 2: Six Histograms  
pdf.savefig(fig3) # Page 3: Placement Rate by Stream  
pdf.savefig(fig4) # Page 4: Correlation Heatmap  
pdf.savefig(fig5) # Page 5: Boxplots
```